ESTIMATION COSTING OF GOYAL HOSPITAL

A project report submitted in partial fulfillment of the requirements for the

7th semester of the degree of

BACHELOR OF TECHNOLOGY

In

ELECTRICAL ENGINEERING

Submitted by

Mahi Kumari (21001007027)

Mannat Raina (21001007028)

Manoj (21001007029)

Under the supervision of

Mr. Satpal Verma

and

Mr. Pawan Kumar

Department of Electrical Engineering

Faculty of Engineering and Technology

J.C. Bose University of Science and Technology, YMCA,

Faridabad, Haryana-121006

December, 2024

CERTIFICATE

This is to certify that Project Report entitled "ESTIMATION AND COSTING OF **"GOYAL HOSPITAL, SEC-08 FARIDABAD**" which is submitted by **MAHI, MANNAT AND MANOJ** in partial fulfillment of the requirements for the award of degree BTech in Department of Electrical Engineering of JC BOSE University of Science & Technology, YMCA Faridabad, is a record of the candidates own work carried out by them under our supervision. The matter embodied in this report is original and has not been submitted for the award of any other degree.

Mr. Satpal Verma

Mr. Pawan Kumar

DECLARATION

We hereby certify that the work which is being presented in this project report entitled **"ESTIMATION COASTING"**, in partial fulfillment of requirements for the 7th semester of degree of BACHELOROF TECHNOLOGY in ELECTRICAL ENGINEERING, submitted to the Department of Electrical Engineering, Faculty of Engineering and Technology, J.C. Bose University of Science and Technology, YMCA, Faridabad, Haryana-121006 is an authentic record of our work carried out during period from July, 2023 to December, 2023, under the supervision of **Mr. Satpal and Mr. Pawan**. We have not submitted the matter presented in this project report to any other University/Institute for the award of B.Tech. or any degree or diploma.

MAHI KUAMRI

(21001007027)

MANNAT RAINA

(21001007028)

MANOJ

(21001007029)

ACKNOWLEDGEMENT

We are overwhelmed in all humbleness and gratefulness to acknowledge our depth to all those who have helped us to put these ideas, well above the level of simplicity and into something concrete. We would like to express our special thanks to our honourable, esteemed guide**, Mr. Satpal Verma** and **Mr. Pawan Kumar** for their able guidance and support in completing our project. His endless patience, guidance, continual encouragement, constant and energetic supervision, constructive criticism, and valuable advice have made it possible to complete this project. Any attempt at any level can't be satisfactorily completed without the support and guidance of our guide. Lastly, we express our heartly regards to our parents whose blessings have always been a continuous source of inspiration to engage ourselves in higher pursuits.

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CHAPTER 1

INTRODUCTION

This report consists of estimation costing done for GOYAL HOSPITAL, SEC-08, PLOT NO-2 OPPOSITE HUMAN TEMPLE, SURDAS PARK, HARYANA 121006.The following are the details of the number of floors and rooms in the hospital

1. UNDERGROUND FLOOR

    - Record room-1
    - Accounts room -1
    - GeneralWard-1
    - LAB-1
    - OPD-1
    - X-ray room-2
    - TPA/ID room-1
    - Washroom-2

2. FIRST FLOOR
    - Male Ward-1
    - Female Ward-1
    - Child Care-1
    - ICU-1
    - Washroom-3

3. GROUND FLOOR
    - Reception-1
    - Waiting room-1
    - No. of rooms-8
    - Dispensay-1
    - No. of Private Rooms-2

4. SECOND FLOOR
    - No of rooms-9
    - No of restricted room-1
    - Operation theatre-1

TYPES OF SUB CIRCUIT

1. Light, fans and 5A socket outlet sub circuit(P<=200W)
2. Sub circuit of 15 A socket point (P<=1000W)

MAIN DISTRIBUTION BOARD

No. of distribution board=3

ASSUMPTION

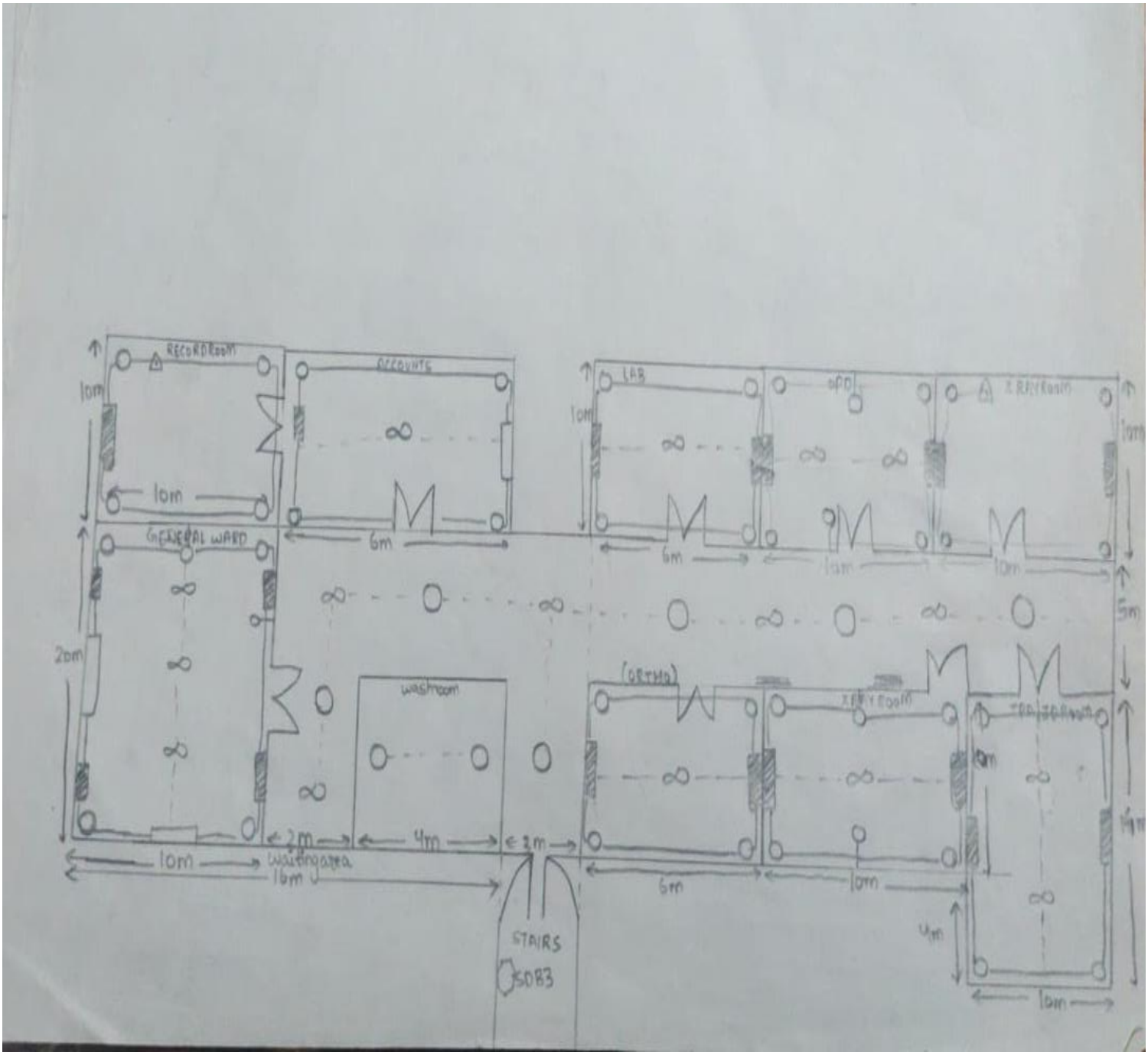
Total height of ceiling=12feet

Fan is hung 1.6 feet below

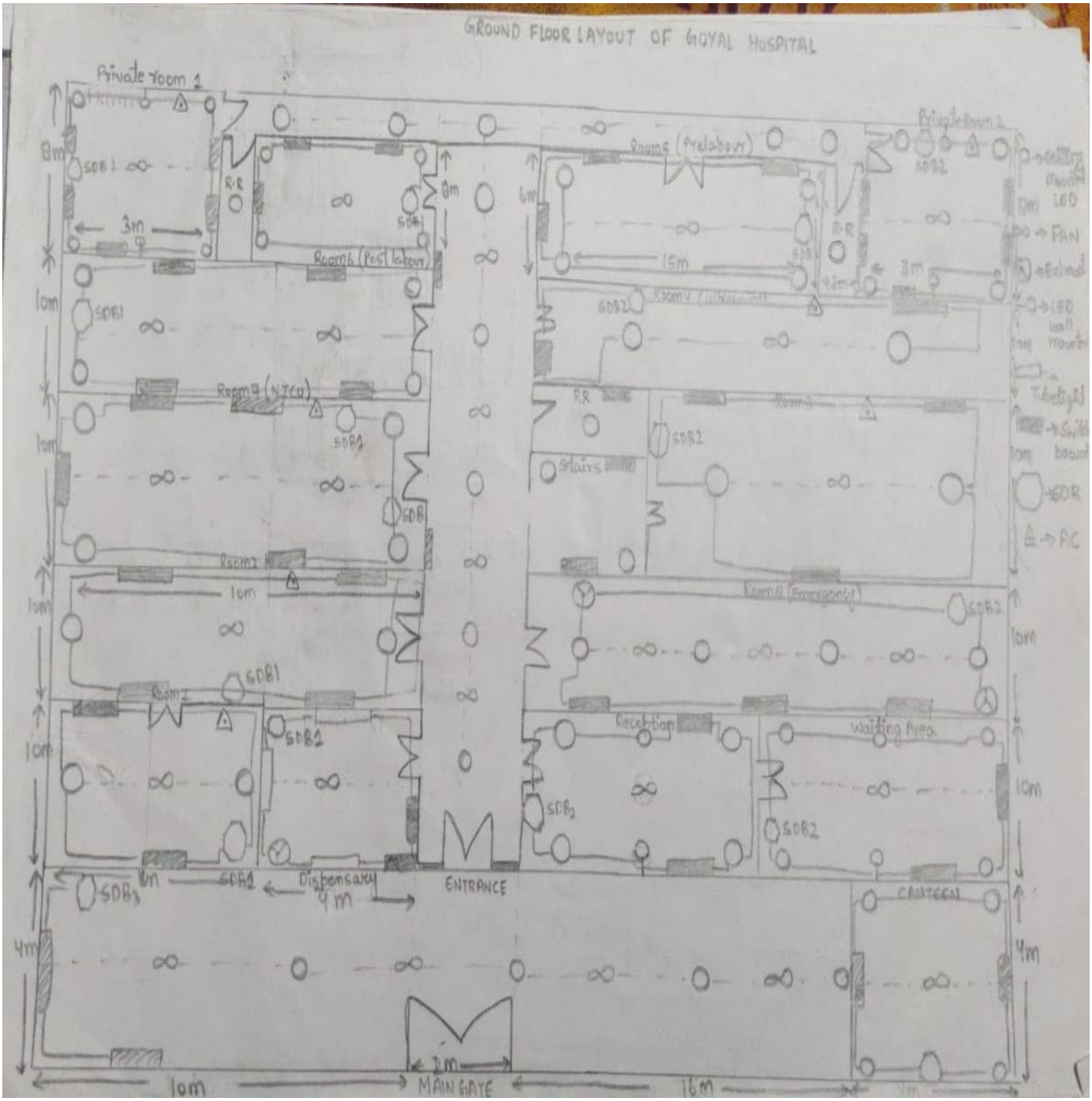
CHAPTER 2

LAYOUT AND WIRING

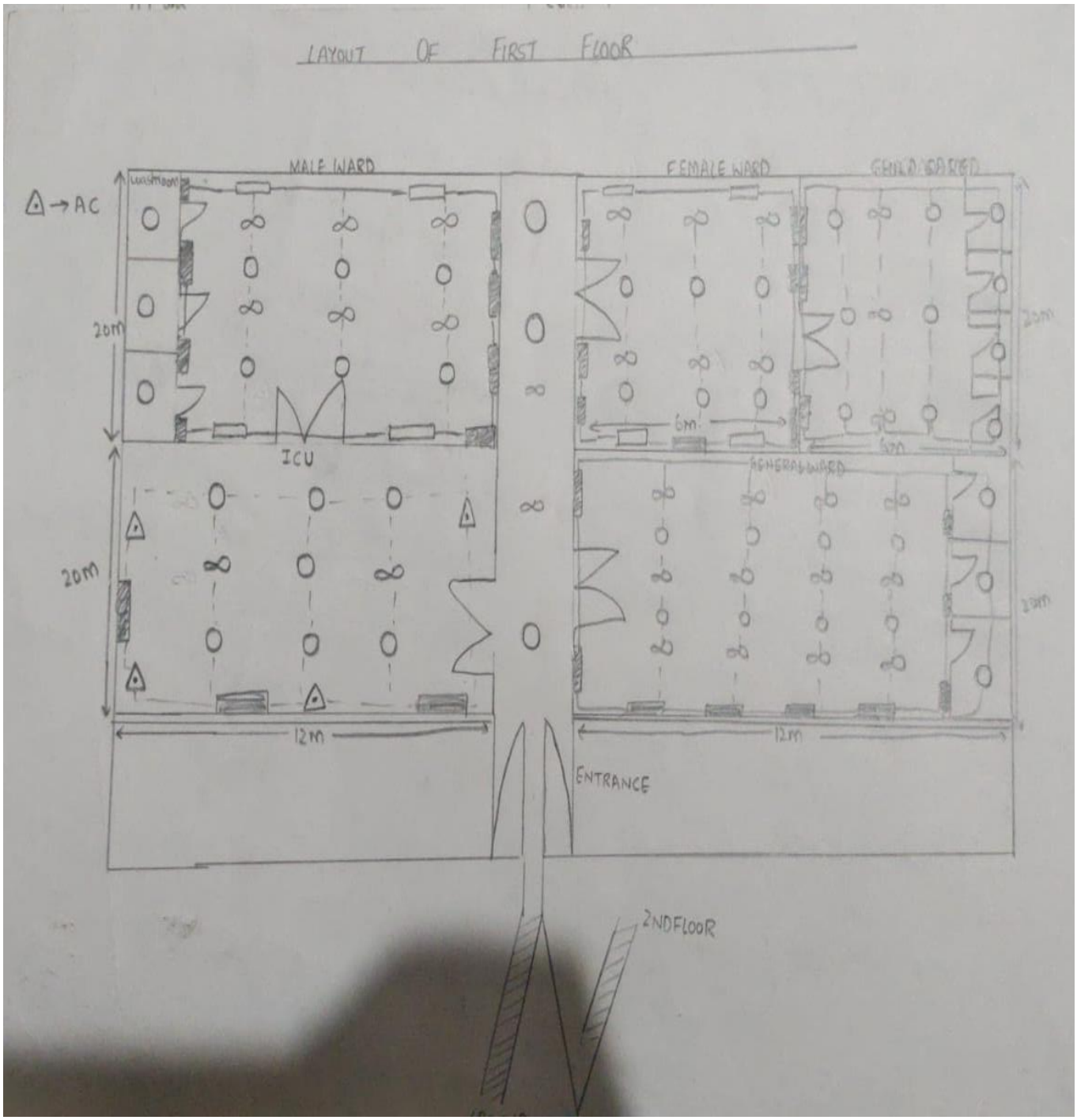
LAYOUT OF UNDERGROUND FLOOR



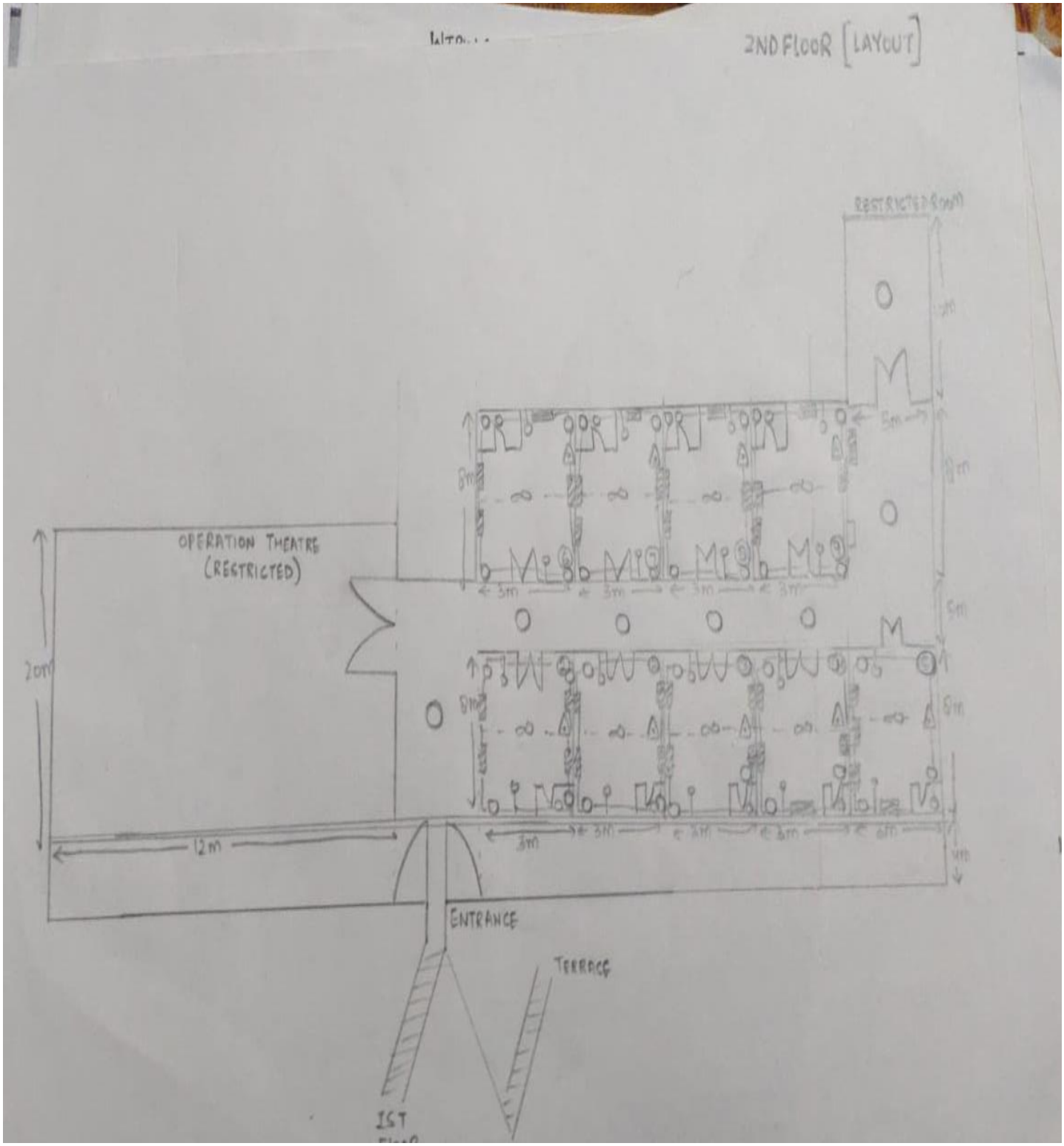
LAYOUT OF GROUND FLOOR



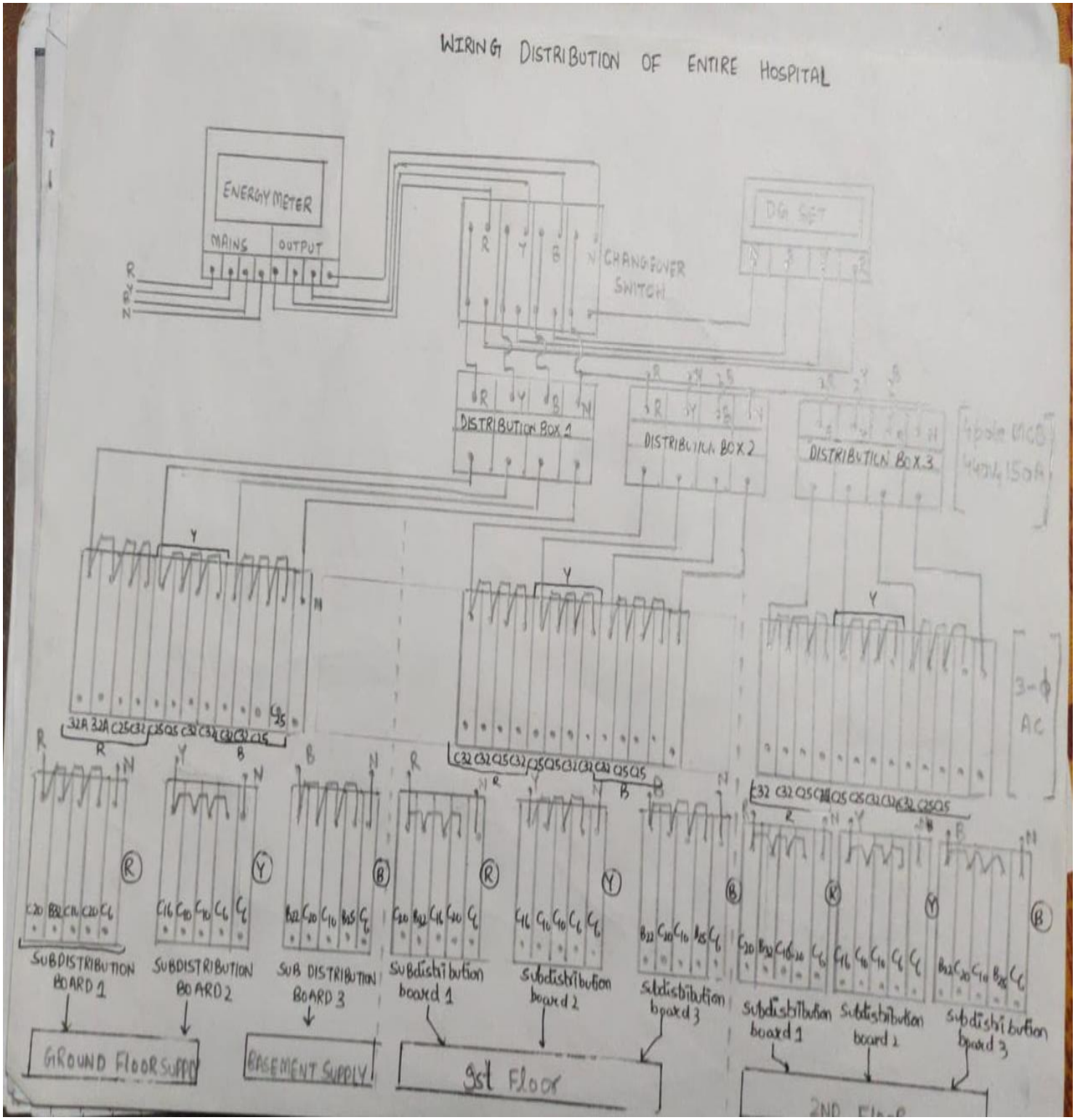
LAYOUT OF 1ST FLOOR



LAYOUT OF 2ND FLOOR



WIRING DISTRIBUTION OF ENTIRE HOSPITAL



CHAPTER 3

LOAD CALCULATION

3.1. LOAD CALCULATION OF UNDERGROUND FLOOR

Total fan points=16 points

Total load= 16*60=960W

Total mounted LED (wall)=6*12=72W

Ceiling LED=42*20=840W

Total 5A socket=49*100=4900W

Total 16 A socket=30*1000W=30000W

Total load=36,772W

Load in Amps=36772 / (230*9) =177.64

Note-Only 60% of the given load is assumed to be connected throughout

Load in amps=0.6*177.64=106.58Amps

3.2. LOAD CALCULATION OF GROUND FLOOR

Fan=28pcs*60=1680W

Ceiling mounted LED=63*20=1260W

Total tube light =3*20=60W

Wall mounted=10*12=120W

5A socket=40*100=4000W

16A socket=50*1000W=50000W

Total=57120W

I(A)275.9A

I(A)=60%=165.5A

3.3. LOAD CALCULATION OF 1ST FLOOR

Total Fan load=31*60w=1860W

Total LED Load =46*20=920W

Total Tube light load=8*20W=160W

Total 5A socket load=77*100=7700W

Total 16A socket load=40*1000W=40000W

Total load in Watts=50640W

Load in Amps=244.63A

Total connected Load in Amps (60%)=146.78A

3.4. LOAD CALCULATION OF 2ND FLOOR

Total LED load=51*20W=1020W

Total Fan load=9*60W=540W

Total LED wall mounted load=18*12=216W

Total 5A socket load=5*100=500W

Total 16A socket load=2*1000W=2000W

Total load=4276W

Total load in Amps=20.675A

Total connected load=12.39A

CHAPTER 4

DETAILED LOAD CALCULATION

4.1. UNDERGROUND FLOOR

1. General Ward

Total fan load=3*60=180W

Total ceiling LED load=4*20=80W

Total LED wall mounted load=2*12=24W

Total tubelight load=2*20=40W

Total 5A laod=10*100=1000W

Total 16 A load=5*1000=5000W

Total load=6324W

2. Record Room

Total ceiling LED load=4*20W=80W

Total 5A laod=2*100=200W

Total 16A load=1*1000=1000W

Total AC load=1000W

Total load=2280W

3. Accounts Room

Total LED Ceiling load=4*20W=80W

Total Fan load=60W

Total tubelight load=20W

Total 5A laod=5*100=500W

Total 16A load=2*1000W=2000W

Total load=2660W

4. Washroom

Total ceiling LED load=2*20W=40W

Geyser point=1*1000W=1000W

Total load= 1040W

5. LAB

Total LED load=4*20=80W

Total fan load=1*60W=60W

Total 5A load=4*100=400W

Total 15A load=2*1000=2000W

Total load=2540W

6. OPD

Total fan load=2*60=120W

Total LED ceiling load=4*20W=80W

Total wall mounted LED load=2*12=24W

Total 5A load=5*100W=500W

Total 16 A load=5*1000W=5000W

Total load=5724W

7. X-RAY room

Total ceiling LED load=8*20W=160W

Total fan load=1*60W=60W

Total AC load=1000W

Total wall mounted LED load=2*12W=24W

Total 5A load=8*100=800W

Total 16A load=8*1000W=8000W

Total load=10044W

8. TPD/ID Room

Total fan load=2*60=120W

Total ceiling LED load=4*20W=80W

Total 5A load=5*100=500W

Total 16 A load=2*1000W=2000W

Total load= 2700W

9. Ortho room

Total fan load=1*60W=60W

Total ceiling mounted LED load=4*20W=80W

Total 5A load=4*100=400W

Total 16 A load=4*1000=4000W

Total load=4540W

10.Corridor

Total fan load=5*60=300W

Total ceiling LED load=6*20=120W

Total 5A load=3*100=300W

Total load =720W

Total Load in wattage of underground floor=38572W

Total load in Amps=186.33A

Total connected load=111.8A

4.2. GROUND FLOOR

1. **Reception**

Total fan load=1*60W=60W

Total ceiling LED load=4*20=80W

Total wall mounted LED load=2*12W=24W

Total AC load=1000W

Socket=10*1000=10000

Total load=17164W

2. **Dispensary**

Total tubelight load=3*20W=60W

Total fan load=1*60W=60W

Total exhaust fan load=1*30W=30W

Load of switch board 1 +switch board 2=

Total load= 2260W

3. **Waiting hall**=200W

4. **Room 1**

Total ceiling mounted LED load=2*20W=40W

Total fan load=1*60=60W

Total 5A load =3*100=300W

Total load =400W

5. **Room 2**

Total fan load =2*60W=120W

Total ceiling LED load=2*20W=40W

Total AC load=1000W

Total 5A load=4*100=400W

Total 16 A load=1000W

Total load=3660W

6. **Balcony**

Total fan load= 5*60=300W

Total ceiling LED load=10*20W=200W

Total 5A load=1*100=100W

Total load =600W

7. **Washroom**

LED wall mounted=2*20W=40W

8. **Room 3**

Total fan load= 1*60W=60W

Total ceiling LED =2*20=40W

Total 5A Load=2*100W=200W

Total 16A Load=2*1000=2000W

Ac load=1000W

Total load= 3300W

9. **Room 4**

Total fan load =60W

Total ceiling LED load= 2*20=40W

Total AC Load=1000W

Total 5A load=300W

Total 16 A load=3000W

10. **Room 5**

Total fan load =60W

Total ac load=1000W

Total 16A load=5000W

Total ceiling mounted led=20*4=80W

Total 5A load=100W

Total load= 6180W

room 5 and room 6 are identical so=6180*2=12360W

11.Room 7

Total fan load=2*60=120W

Total ac load=1000W

Total ceiling LED load=4*20W=80W

Total 5A load=2*100W=200W

Total 16A load=15*1000=15000W

Total load= 16600 W

12.Room 8

Total load=1040W

4.3. 1ST FLOOR

1. Male ward

Total fan load=6*60=360W

Total ceiling LED=9*20=180W

Total tube light load=4*20W=80W

Total 5A load=16*100=1600W

Total 16A load=8*1000=8000W

Total load=10,220W

2. ICU

Total AC load=4000W

Total fan load=120W

Total 5A load=1200W

Total 16A load=12000W

Total LED load=140W

Total load=17460W

3. Female ward

Total fan load=6*60W=360W

Total LED load=6*20=120W

Total tube light load=4*20=80W

Total 5A load=1600W

Total 16 A load=8000W

Total load=10160W

4. General ward

Total fan load=12*60=720W

Total LED load=11*20=220W

Total 5A load=27*100=2700W

Total 16A load=9*1000=9000W

Total load=12640W

5. Child care ward

Total LED load=10*20=200W

Total fan load=3*60=180W

Total 5A load=600W

Total 16A load=3000W

Total load=3980W

6. Lobby

Total fan load=2*60W=120W

Total LED load=3*20=60W

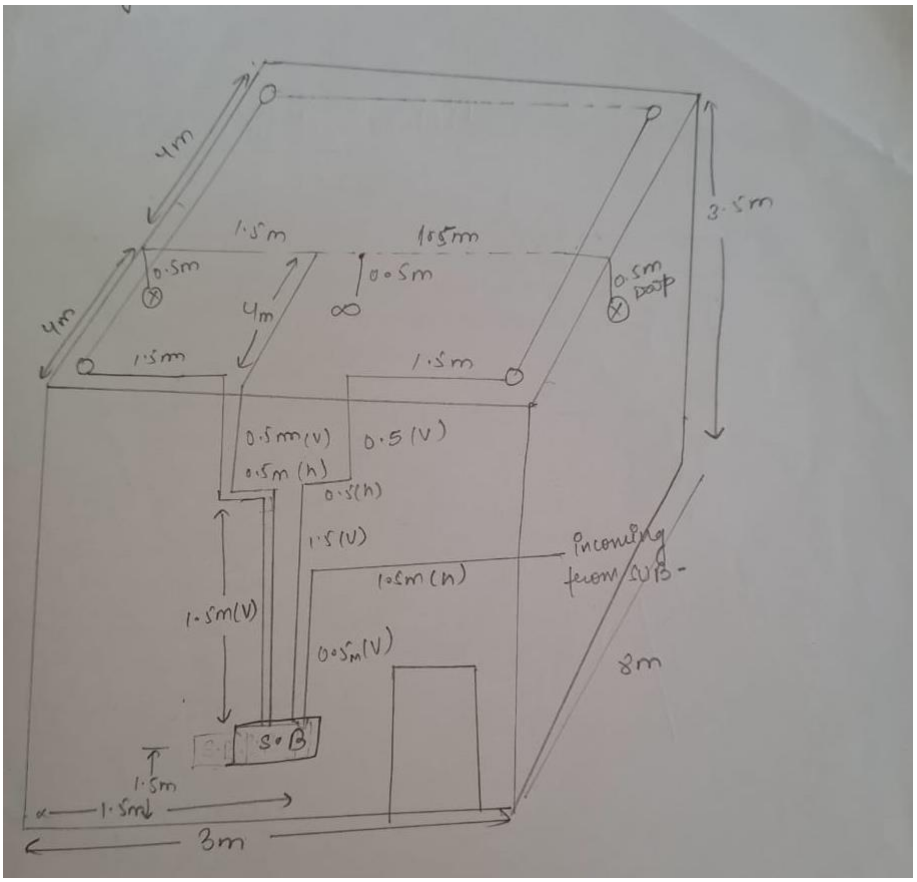
Total load=180W

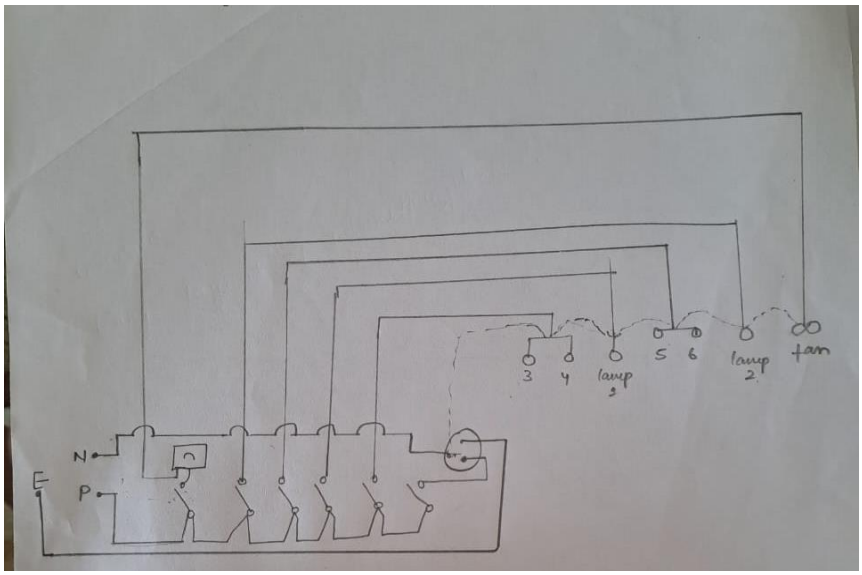
Total load of this floor=54640W

Load in Amps= 263.96A

Total connected load=158.3A

Wiring Calculation for 1ST Floor





Calculation of length of conduit

h= horizontal, v=vertical, c=ceiling

from switch board to fan=1.5(v)+0.5(h)+0.5(v)+4=6.5m

from fan to lamp1=1.5(h)+0.5(v)=2m

from fan to lamp 2=1.5(h)+0.5(v)=2m

from switch board to lamp 3 =1.5+0.5+0.5+1.5=4.0m

from switch board to lamp 4=4m

lamp 3 to lamp 5 =8m

lamp 4 to lamp 6=8m

total length of conduit=37.5m

taking 10% wastage =3.75m

total length of conduit pipe=41.25m

Calculation of length of phase wire

from incoming to switch board =1.5(h)+0.5(v)=2m

from switch board to fan= 1.5(h)+0.5(h)+0.5(v)+4(c)=0.5(v)=7m

from switch board to lamp1=1.5+0.5+0.5+4+1.5+0.5=8.5

from switch board to lamp2=1.5+0.5+0.5+4+1.5+0.5=8.5

from switch board to lamp3=1.5+0.5+0.5+4+1.5=8

from switch board to lamp4=1.5+0.5+0.5+4+1.5=8

total length=42m

taking 10% wastage =4.2m

total length of phase wire=46.2m

Calculation of neutral wire

from incoming to switch board=1.5+0.5=2m

switch board to fan+1.5+0.5+0.5+4+0.5=7m

fan to lamp 1=1.5+0.5=2m

fan to lamp 2=2m

lamp 3 +lamp 4=8m

lamp 5+ 6=8m

total length of neutral wire= 29 m

taking 10% wastage =2.9m

total length of neutral wire=31.9m

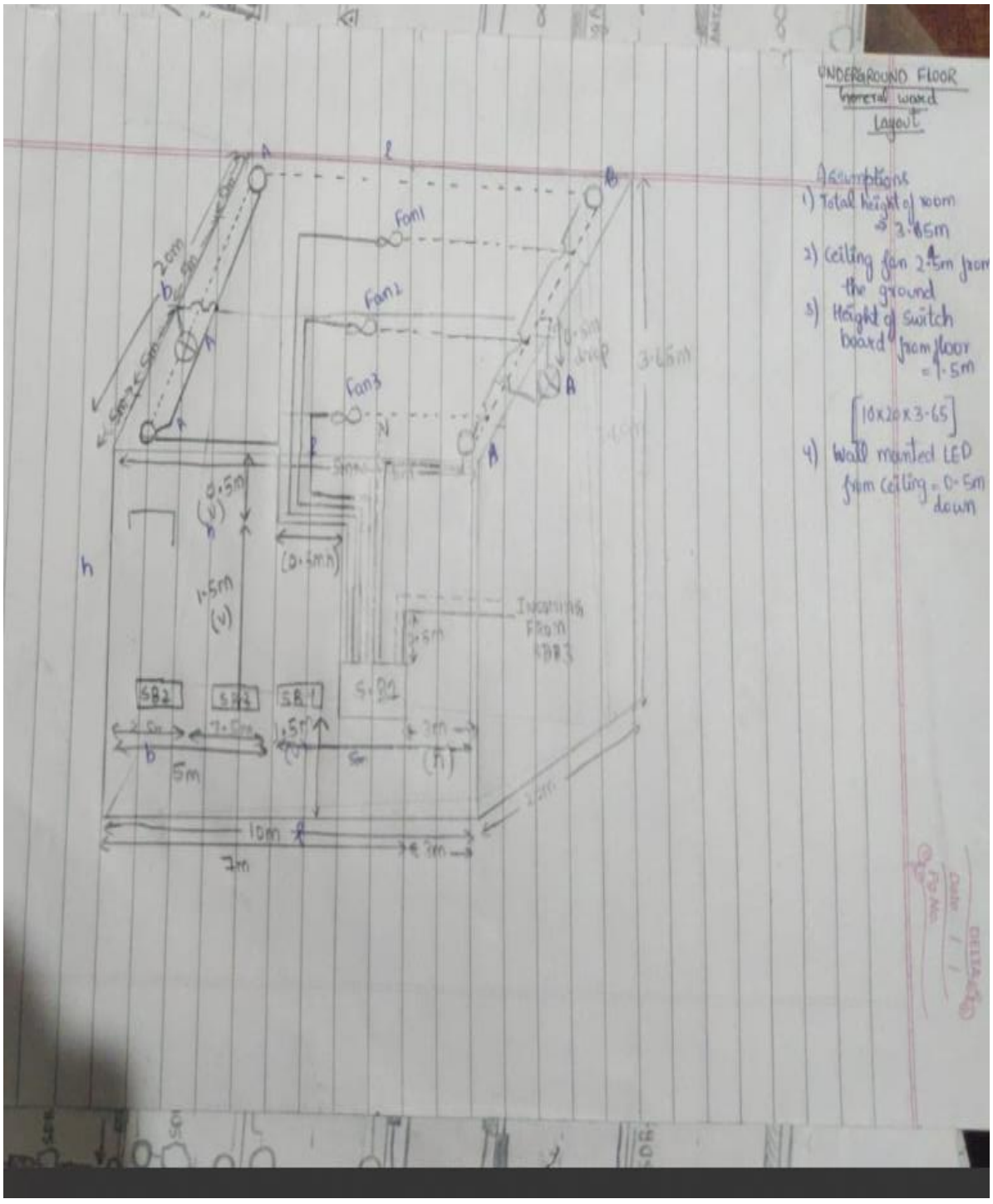
Note:- Second floor consists of 10 rooms

so total length of phase wire=462m

total length of neutral wire =319m

wire size used 1.5mm diameter

Wiring calcualtion for underground floor



Calculation of length of phase wire(sub-circuit)

From sub distribution board to switch board1=3m(h)+0.5m=3.5m

From switch board 1 to fan 1= 1.5(v)+0.5(v)+0.5(h)+5m=7.5m

From switch board to fan 2=12.5m

From switchboard 1 to fan 3=17.5m

From switch board 1 to led A=1.5(v)+0.5(h)+0.5(v)+4.5(H)+20m=27.5m

From switch board to led B=27.5m

Total length of phase wire=96m

Total Phase Wire Length (for 8 rooms):

96m × 8 = 768m for all 8 rooms.

Including 10% Wastage:

Wastage = 10% of 768m = 76.8m

768m + 76.8m = 844.8m = rounded to 845m.

Note: 1.5mm2 wire is used entirely

For sockets (5A&16A)

From sub distribution board 3 to switch board2,3 and 4

Total =3.5m

For 16A =2.5mm2 wire is used

For 5A =1.5mm2 wire is used

Calculation of Conduit

From Switch Board 1 to fan 1+fan2+fan

3=1.5(v)+0.5(h)+0.5(v)+16(h)+5(h)=23.5m

Length of SB1 to LED's A =2.5+4.5+20.5=27.5m

From SB1 to LED's B=2.5+5.5+20+0.5=28.5m

From Sub distribution board 3 to switch board1=3.5m=83m

Total Conduit Length (for 8 rooms):

83m × 8 = 664m for all 8 rooms.

Including 10% Wastage:

Wastage = 10% of 664m = 66.4m

664m + 66.4m = 730.4m=rounded to 730m.

Calculation of neutral wire

From SB1 to fan 1,2,3=22.5m

From SB1 to LED A=0,5m+5m+20m=22.5m

From SB1 to LED B=19m

Total=64m

Total Neutral Wire Length (for 8 rooms):

64m × 8 = 512m for all 8 rooms.

Including 10% Wastage:

Wastage = 10% of 512m = 51.2m

512m + 51.2m = 563.2m, rounded to 563m.

Wire used is 1.5mm2

Wiring calculation of 2ND floor

Phase wire Calculation

From Sub Distribution Board to Switch Board 1
4m (h) + 0.5m = 4.5m

From Switch Board 1 to Fan

2m (v) + 1m (v) + 1m (h) + 6m = 10m

From Switch Board 1 to Fan 2
14m

From Switch Board 1 to Fan 3:
18m

From Switch Board 1 to LED A:
1.5m (v) + 0.5m (h) + 0.5m (v) + 6m (h) + 22m = 30m

From Switch Board 1 to LED B :
30m

Total Length of Phase Wire (per room):

Total = 4.5m (Sub DB to SB1) + 10m (SB1 to Fan 1) + 14m (SB1 to Fan 2) + 18m (SB1 to Fan 3) + 30m (SB1 to LED A) + 30m (SB1 to LED B)

Total = 106.5m per room.

Note: 1st floor consists of 7 rooms

106.5m × 7 = 745.5m

Add 10% wastage:

Wastage = 10% of 745.5m = 74.55m

745.5m + 74.55m = 820.05m, rounded to 820m.

Wire Type:

1.5mm² wire is used for all phase wiring.

Conduit Calculation

From Switch Board 1 to Fans 1, 2, and 3 (per room): 2m (v) + 1m (h) + 1m (v) + 20m (h) + 7m (h) = 31m

From Switch Board 1 to LED
3m (v) + 7m (h) + 23m (h) = 33m

From Switch Board 1 to LED B
3m (v) + 6m (h) + 22m (h) + 1m (h) = 32m

From Sub Distribution Board to Switch Board 1:
4.5m

Total Length of Conduit

Total = 31m (for fans) + 33m (for LED A) + 32m (for LED B) + 4.5m (from Sub DB to SB1)

Total = 100.5m.

Now, multiply by 7 rooms:

100.5m × 7 = 703.5m

Add 10% wastage:

Wastage = 10% of 703.5m = 70.35m

703.5m + 70.35m = 773.85m, rounded to 774m.

Neutral Wire Calculation :

From SB1 to Fans 1, 2, and 3
23m

From SB1 to LED A
1m (v) + 6m (h) + 22m (h) = 29m

From SB1 to LED B (per room):
25m

Total Length of Neutral Wire

Total = 23m (for fans) + 29m (for LED A) + 25m (for LED B) = 77m per room.

77m × 7 = 539m for all 7 rooms.

Adding 10% Wastage:

Wastage = 10% of 539m = 53.9m

539m + 53.9m = 592.9m, rounded to 593m

Wire Size: 1.5mm²

Wiring Calculation for Ground Floor

Phase wire calculation

From Sub Distribution Board to Switch Board 1
4.5m (h)

From Switch Board 1 to Fan 1
2m (v) + 1m (v) + 1m (h) + 6m = 10m

From Switch Board 1 to Fan 2
14m

From Switch Board 1 to Fan 3
18m

From Switch Board 1 to LED A
1.5m (v) + 0.5m (h) + 0.5m (v) + 6m (h) + 22m = 30m

From Switch Board 1 to LED B
30m

Total Length of Phase Wire

Total = 4.5m (Sub DB to SB1) + 10m (SB1 to Fan 1) + 14m (SB1 to Fan 2) + 18m (SB1 to Fan 3) + 30m (SB1 to LED A) + 30m (SB1 to LED B)

Total = 106.5m per room.

Note : it consists of 10 rooms

106.5m × 10 = 1065m for all 10 rooms.

considering 10% wastage:

Wastage = 10% of 1065m = 106.5m

1065m + 106.5m = 1171.5m, rounded to 1172m.

Total Phase Wire Length (with 10% Wastage): 1172m

Wire Size: 1.5mm²

Ground Floor: Conduit Calculation

From Switch Board 1 to Fans 1, 2, and 3 (per room): 2m (v) + 1m (h) + 1m (v) + 20m (h) + 7m (h) = 31m

From Switch Board 1 to LED A
3m (v) + 7m (h) + 23m (h) = 33m

From Switch Board 1 to LED B
3m (v) + 6m (h) + 22m (h) + 1m (h) = 32m

From Sub Distribution Board to Switch Board 1 (per room):
4.5m

Total Length of Conduit

Total = 31m (for fans) + 33m (for LED A) + 32m (for LED B) + 4.5m (from Sub DB to SB1)

Total = 100.5m per room.

Now multiply by 10 rooms:

100.5m × 10 = 1005m for all 10 rooms.

Add 10% wastage:

Wastage = 10% of 1005m = 100.5m

1005m + 100.5m = 1105.5m= rounded to 1106m.

Final Conduit Calculation for Ground Floor:

Total Conduit Length (with 10% Wastage): 1106m

Neutral Wire Calculation:

From SB1 to Fans 1, 2, and 3 (per room):
23m (for all fans combined per room)

From SB1 to LED A (per room):
1m (v) + 6m (h) + 22m (h) = 29m

From SB1 to LED B =25m

Total Length of Neutral Wire

Total = 23m (for fans) + 29m (for LED A) + 25m (for LED B) = 77m per room.

77m × 10 = 770m for all 10 rooms.

Adding 10% Wastage:

Wastage = 10% of 770m = 77m

770m + 77m = 847m

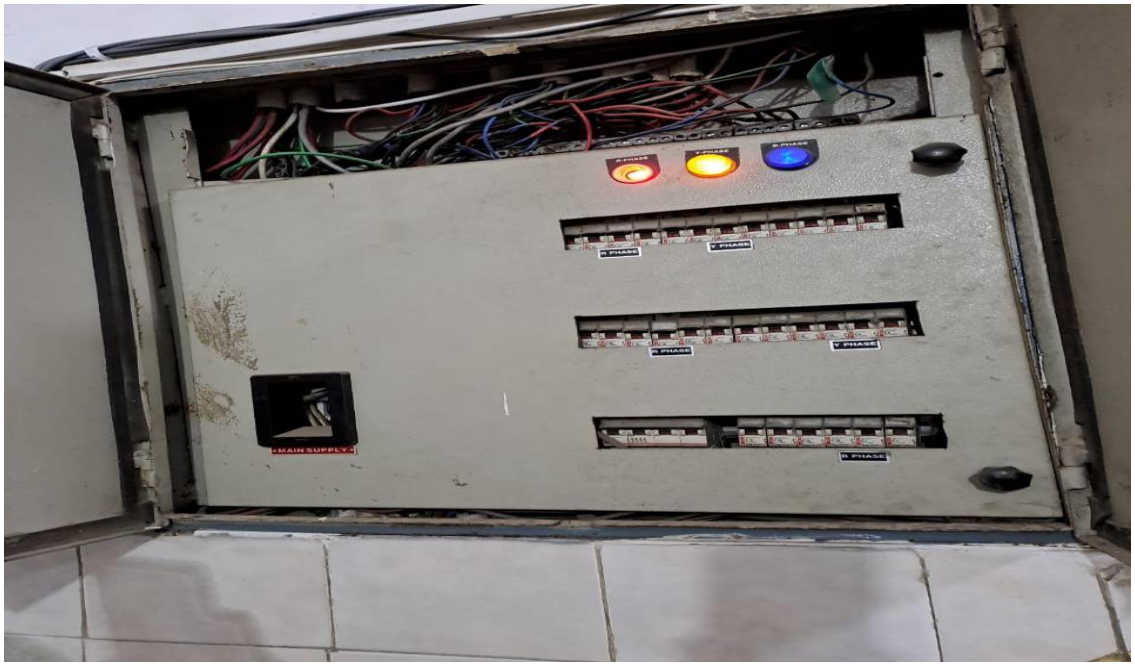
Wire size=1.5mm2

Material list

S.NO	Specifications	Quantity	Cost per Article (Rs.)	Cost(Rs.)
1.	MCB Main switch 440V,200A A rating	1	10000	10000
2.	MCB 32 A rating	15	500	7500
3.	MCB 20A rating	15	300	4500
4.	MCB 16A	20	250	5000
5.	MCB 6A	24	150	3600
6.	MCB 25A	15	350	5250
7.	Conduit pipe 25mm	3200m	60/m	192000
8.	Bakelite sheet(10*10cm)	100	50	5000
9.	Bakelite sheet(20*10cm)	150	80	12000
10.	Fan box	200	120	24000
11.	Conduit junction box	150	150	22500
12.	5A switch (modular)	200	150	30000
13.	5A socket(modular)	150	200	30000
14.	16A Switch (modular)	220	300	66000
15.	16A socket (Modular)	150	300	45000
16.	Lamp holder	50	80	4000
17.	Electric tape	50	50	2500
18.	Ceiling roses	100	150	15000
19.	PVC 2.5 mm wire(Polycab)	500m	2917/roll(90m)	17502
20.	PVC 1.5 mm wire	5621m	1370/roll(100m)	78090
21.	PVC 4 mm wire	800m	4085/roll(90m)	36765
22.	Energy meter	1	2500	2500
23.	Conduit bends	300	50	15000
24.	Fan regulator	100	250	25000
25.	Isolator	1	1500	1500
26.	ELCB	1	4000	4000
27.	Miscellaneous expenses@10% of total cost			50000
	TOTAL COST OF WIRING OF BUILDING			7,14,207








R PHASE
Y PHASE
B PHASE
R PHASE


GOYAL HOSPITAL
गोयल अस्पताल
EMERGENCY 24/7
+91 93122 94755
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